

Otsu Thresholding:

Defn.:

- Automatic Thresholding system
- Converts Grayscale image into a binary image by choosing the optimal threshold based on pixel intensity

Purpose:

- Automatically separates *foreground* and *background*
- No need to use manual thresholding selection

Application:

- Image Segmentation
- Medical Image Analysis
- Document binarization (text extraction)
- Preprocessing for:
 - Edge detection
 - Object detection
 - Computer Vision pipelines

Example:

- Image:

10	12	15	200	210
12	14	18	205	220
11	13	16	198	215
9	10	14	202	225
8	11	13	207	230

Observation:

Left side → dark pixels (background)

Right side → bright pixels (foreground)

- Otsu will find a threshold somewhere around ~100-150 to separate these two groups

Key Concepts:

- Maximize between-class variance:

$$^2\sigma_b = w_0 w_1 (\mu_0 - \mu_1)^2$$

Algorithm(Pseudocode):

INPUT: grayscale image I (values 0-255)

1. Compute histogram hist[0...255]
For each pixel value v in image:
 hist[v] += 1
2. Normalize histogram to probabilities p[i]
total_pixels = total number of pixels
For i = 0 to 255:
 p[i] = hist[i] / total_pixels
3. Compute total mean intensity μ_{total}
 $\mu_{total} = \sum (i * p[i])$ for i = 0 to 255
4. Initialize:
 w0 = 0
 μ_0 = 0
 max_variance = 0

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best_threshold = 0
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5. For each threshold  $T$  from 0 to 255:
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     $w0 = w0 + p[T]$   
     $\mu0 = \mu0 + T * p[T]$ 
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    If  $w0 == 0$ :  
        continue
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     $w1 = 1 - w0$   
    If  $w1 == 0$ :  
        break
```

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     $\mu0\_current = \mu0 / w0$   
     $\mu1 = (\mu\_total - \mu0) / w1$ 
```

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     $\sigma\_b^2 = w0 * w1 * (\mu0\_current - \mu1)^2$ 
```

```
    If  $\sigma\_b^2 > max\_variance$ :  
         $max\_variance = \sigma\_b^2$   
         $best\_threshold = T$ 
```

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6. Return best_threshold
```

Final Thresholding:

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If pixel  $\geq T \rightarrow 255$ 
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Else  $\rightarrow 0$ 
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